

AI4GA Learning Objectives (v.2_{updated 8/23/2022})

Unit 1 Modules

1.1 - Living and Working with Artificial Intelligence

Students should know

EU.1.1.1 The Five Big Ideas in AI

Students should be able to:

LO.1.1.1 Describe AI to someone

LO.1.1.2 Give examples of how they are affected by AI technology in their own daily lives

1.2 - Autonomous Robots

Students should be able to:

LO.1.2.1 Describe the characteristics of a “robot”

LO.1.2.2 Define “autonomous”

LO.1.2.3 Describe the kinds of decisions a given robot makes

LO.1.2.4 Describe at least three different uses, functionalities, and benefits of autonomous robots

LO.1.2.5 **Describe the kinds of things autonomous robots look for**

LO.1.2.6 List different types of workplaces that can use autonomous robots

LO.1.2.7 Describe careers in AI and Robotics

1.3 - Anatomy of Self-Driving Vehicles

Students should know that

EU.1.3.1 Self-driving cars are a type of autonomous robot

Students should be able to:

LO.1.3.1 Describe how self-driving cars work

LO.1.3.2 Illustrate and describe the sensors commonly found in/on a self-driving car including sensor placement, usage, how it works.

LO.1.3.3 Describe the things self-driving cars robots look for

LO.1.3.4 Illustrate and describe the subsystems commonly found in self-driving cars (e.g., perception, localization, navigation (route planning), maneuver planning, vehicle control, collision avoidance)

LO.1.3.5 Describe the impacts self-driving vehicles (e.g., cars, trucks, delivery robots) have on society

1.4 - Robot & Computer Perception (mechanisms of perception - how it makes meaning)

Students should know that

- EU.1.4.1 An edge is a significant change in brightness along a boundary in an image.
- EU.1.4.2 Edge detection is the process of finding edges in an image. It is an initial step for finding more complex features in images.
- EU.1.4.3 Edge detection is used in facial recognition to find the shapes of the mouth, eyes, nose, and jawline. In self-driving cars it is used to help find street signs, lane markers, and people.
- EU.1.4.4 Computer vision is a discipline within computer science concerned with getting computers to see (perceive) things. This includes recognizing faces, body poses, objects, scenes, and handwriting, and even interpreting X-rays and MRI scans.

Students should be able to:

- LO.1.4.1 Identify applications of face detection in everyday life
- LO.1.4.2 Create a digital artifact using face detection technology
- LO.1.4.3 Explain how face filters apply graphics to an image
- LO.1.4.4 Identify applications of computer vision in everyday life
- LO.1.4.5 Identify applications of object detection in self-driving cars and autonomous robots
 - a. Ex. Street signs, traffic cones, traffic lights, people (e.g., pedestrians, people directing traffic), lane markings

1.5 Route Finding

Students should be able to:

- LO.1.5.1 Create a graph representation of a map
- LO.1.5.2 Describe (verbalize and illustrate) how to find a route between two locations by graph coloring
- LO.1.5.3 Describe (verbalize and illustrate) how to find a route between two locations by drawing a search tree
- LO.1.5.4 Construct a search tree using breadth-first search (Foreshadowing future modules)

1.6 - Case Study

Students should be able to:

- LO.1.6.1. Describe social and ethical issues of autonomous vehicles
- LO.1.6.2. Describe the benefits and harmful impacts of autonomous vehicles
- LO.1.6.3. Evaluate a manufacturer's instructions for operating a car to determine the car's level of autonomy
- LO.1.6.4. Consider multiple perspectives when evaluating AI technology

1.7 - Mini-Projects

LO.1.7.1. Adf

1.8 Dreambot

- LO.1.2.1 Describe the characteristics of a “robot”
- LO.1.2.3 Describe the kinds of decisions a robot makes

- L.O.1.D.1. Design a robot that solves a problem in everyday life
- L.O.1.D.2. Describe how robots work
- L.O.1.D.3. Illustrate and describe the sensors commonly found in/on a robot including sensor placement and their usage.
- L.O.1.D.4. Illustrate and describe the subsystems commonly found in a robot (e.g., perception, localization, navigation (route planning), maneuver planning, vehicle control, collision avoidance)
- L.O.1.D.5. Describe the positive and negative impacts robots (e.g., cars, trucks, delivery robots) can have on society

Extension Content Deeper Dive

Links to module content

[Module 1.6 - Shared Deep Dive: Cars and AI Driver Assistance](#)

[Module 1.6 - Shared Deep Dive: Businesses, robots, and autonomous vehicles](#)

<https://docs.google.com/document/d/1Bp43rb9cljxJ0v87v7yLQky-pEt0CErImVKb5X7L6A/edit?usp=sharing>

- LO.1.E.1 Describe the benefits and harmful impacts of autonomous vehicles

Unit 2 Modules

2.1 - Understanding Language Overview

Students should know that

- Language has evolved as ... culture
- NLP
- Knowledge that agents are using to respond to speech

Students should be able to:

- Distinguish between sensing and perception
- Language
- Identify applications of speech recognition or audio-based recognition in the real-world
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2.2 - How do computers understand what we say?

Students should be able to:

- Explain how speech recognition works
- Explain the nature of sound and how sound waves are visualized (oscilloscope)
- Explain how different pitch sounds can be discriminated in a spectrogram
- Use spectrogram tool to distinguish high and low pitch sounds
 - Read spectrogram
- Explain how we can tell different speech sounds apart using the spectrogram
- Microphone is a sensor that generates a electrical signal that is visualized/graphed by an oscilloscope. The spectrum analyzer converts from a waveform to a spectrogram which necessary for discriminating different sounds.
 - slide : waveform of aah, eee, ooo (similar and non-discriminable), spectrogram (show different energies)
 - If you have more than 3
 - The spectrogram is a more reliable way of making the distinction.

2.3 - How do intelligent assistants understand and answer questions?

Students should know

- Question Answering
- Representation for the questions - Berkeley Neural Parser
- Parts of speech: nouns, verbs, prepositions
- Syntactic relationships
- How BNP works - Not a purely syntactic parser it uses meaning of words and statistics
- Semantic parsing

- Sources of knowledge
- Limitation: Systems are a collection of techniques

Students should be able to:

- Identify areas of ambiguity in human speech that makes it complex
- Explain how computers/AI overcome the ambiguities in human speech

2.4 - Word Embeddings

Students should be able to:

- Illustrate abstraction hierarchies for speech understanding to highlight the relationship between higher level and lower levels of abstraction in resolving ambiguities in the levels
- Explain why AI has challenges understanding stories, human, sarcasm, etc

2.5 - How Computers Represent and Generate Meaning

Students should be able to:

- Explain how computers can generate meanings (machine translation)
- Explain the limitations of language translation systems
- Describe what are transformer neural networks used for

2.6 - Sentiment Analysis

Students should be able to:

- Define what Natural language processing is
- Describe the difference between subjectivity and polarity in sentiment analysis
- Describe limitations of sentiment analysis tools

2.7 - Chatbots

Students should be able to:

- Explain how semantic information webs capture meaningful relationships between entities
- Identify the limitations of chatbots

2.8 - Case Study

- N/A

2.9 - Mini-Project

- N/A

Unit 3 Modules

Betia - We put a bunch of notes throughout this document to help explain what resources to use as you develop the online PD.

3.1 - How do computers make decisions?

- Identify the kinds of decisions that computers make (i.e., classification, prediction, recommendation, planning & scheduling)
- Define reasoner
- Demonstrate how computers make different kinds of decisions in everyday life

3.2 - How can a computer learn to classify objects from examples? (Old unit 3 deck module 3.3)

Students should be able to:

- Create a decision tree classifier
- Use a decision tree to classify objects
- Inventing a set of features that capture the differences between classes
- Create feature vector representations
- Evaluate the effectiveness of a model or reasoner
- Define Machine Learning
- Explore different types of Machine Learning*

Note to Betia - Teacher PD - Slides - 5-7 - create a decision tree classifier or Guess the animal slides -  Guess the Animal

* indicates LOs only for teachers not for students

** Don't include Pasta Land in the online PD we want to do this as an interactive activity in the July PD.

*** Please use the slides from the old Year 1, Unit 3 slide deck. It captures more of the points needed to teach the concepts so that the teachers have a better idea of the nuances of the activities.

Pasta land Activity**

- Creating a decision tree classifier to recognize different types of pasta.
 - Inventing a set of features
 - Growing the decision tree one node at a time
- Use a decision tree to classify new pasta examples

- Assess whether the current feature set is sufficient
- Create feature vector representations for the different pasta types

Takeaway

- Define Machine Learning
- Explore different types of Machine Learning

Vocabulary

- Features
- Feature Vectors

3.3 - How do we train a computer to make decisions (Old unit 3 deck module 3.4)

- Construct a decision tree to solve a classification problem using a software tool
- Demonstrate how a computer learns to classify inputs from examples
 - Understand the algorithm is following this process:
 - What is the feature that will help me make progress in getting the right answers? What am I still getting wrong? (repeat)
 - Understand that sometimes a single feature is enough to solve the problem, but usually it is not and more features are needed.
- Automated procedures for making decision tree
- Learn a class from examples
 - Providing them the Fish Feature space & examples
- Explain how labeled datasets are used to train a classifier (e.g., classification reasoner)

3.4 - How can we use Machine Learning to answer questions with datasets? (Working with Tabular Data) (Old unit 3 deck module 3.5)

- Describe the stages of the machine learning pipeline for training a model
 - Slides 5, 7, 8, 12, 12 [\(old module 3.5\)](#)
- Describe the different types of data
 - Slide 22 [\(old module 3.5\)](#)
- Explain why a dataset is split into two sets: a training and a test set
 - Slide 25 [\(old module 3.5\)](#)
- Train a classification or prediction model by selecting features in a dataset (feature engineering)
- Explain how models are trained using labeled data

- A ML algorithm is used to find patterns in the training data.
 - Patterns are statistical relationships called correlations between the value to be predicted and the features selected for the training.
- Distinguish between training and using a model
 - Training a model is when the computer is learning
 - Using a model/reasoner is when learning has stopped and we have a finished model that we apply to new problems
- Test and evaluate the accuracy of the model
- Explain why results of models trained on the same dataset may be different
 - Your results may be different even though you picked the same features because of how the training and test sets are randomly split.

Vocabulary

- Tabular dataset
- Dataset - slide 9
- Data
- label
- Pipeline
- Feature engineering

3.6 - Neural Networks

- Explain how neural networks work
 - Slide 7  Module 3.6 - Neural Networks
- Describe the general architecture (i.e., organization) of a neural network and its components
 - Components: Neurons, layers, inputs and outputs
 - Neural networks are generally organized into input layer, hidden layer(s), and output layer. Each layer is made up of neurons. Neurons in the same layer are not connected. Rather the neurons in one layer are connected to the neurons in the layer next to them.
 - [Slides 8-9]  Module 3.6 - Neural Networks
- Explain how neurons learn using supervised learning
 - Dog Bites example Slides 11-28  Module 3.6 - Neural Networks

Note to Betia - Don't worry about doing how neurons compute their answer - slides 30+. Dave has been developing a new tool and we will use it during the online PD in July

3.7 - Case Study: Does AI Make Better Decisions than People?

- Identify current uses of AI to make decisions and how it has impacted society or people
- Evaluate the ways various stakeholders' values influence the design of an AI System
- Identify and explain sources of bias in an AI system and decision making process

- Explain how and why issues of bias, fairness, and transparency affect the design and use of AI systems and impacts on people
- Create example laws, policies, and procedures that regulate the design and use of AI Systems that make decisions about people.

3.8 - Mini-Projects

- Create a classifier using text, numbers, images, audio, or pose
 - Slide 6  Module 3.8 - Mini-Project Overview
- Create a classifier using an age and skill appropriate ML tool
- Describe the ethical responsibilities of designing an AI model or system
 - Slide 2  Module 3.8 - Mini-Project Overview
- Explain the types of bias in ML
- Explain the role of model cards and the types of information it communicates
 - Slide 4  Module 3.8 - Mini-Project Overview
- Create a Model Card - describing the intended use of your system, source of data, limitations, and warnings.
- Evaluate the potential positive and negative an AI system can have users
- Explain the steps taken to reduce and limit bias in a dataset and process of training a model

AI4GA Learning Objectives (v.1)

Notes about removals and comments from Previous Draft that still need to be addressed

Formerly Unit 1 Module 1.1

- ~~LO.1.1.3 — Give examples of what AI looks like in their community~~
 - 1.1.o. Hard to answer this
 - 1.1.o. License plate readers
 - 1.1.o. Uses in industry
 - 1.1.o. Another option - Activity - Imagine what AI might look in our community (should this be about solving a problem)
 - 1.1.o. Missing - Stakeholder objective
 - 1.1.o. (Recurring LO for each unit - Move to the end of unit and course)
 - 1.1.■. NEW LO - Apply what you know about AI to
 - 1.1.■. New LO - Describe possible/plausible uses of AI technology in their home, school, or community
 - 1.1.●. E.g., deer detector - to prevent accidents and animal

Formerly Unit 1, Module 1.2

- LO.1.2.4 Describe at least three different uses, functionalities, and benefits of autonomous vehicles/robots
- LO.1.2.5 List several industries that use autonomous robots
- LO.1.2.6 Describe the impacts autonomous robots have on society
- ~~LO.1.2.7 — Describe the kinds of problems robots solve~~
 - ~~LO.1.2.a — All of these robots are solving the same kinds of problems. How does each robot solve them?~~
 - ~~LO.1.2.i — Where am I?~~
 - ~~LO.1.2.ii — Where do I want to be?~~
 - ~~LO.1.2.iii — How do I stay safe? (obstacles, humans, cliffs, boundaries)~~

Formerly Unit 1, Module 1.3

- LO.1.3.6 Illustrate and describe the subsystems commonly found in self-driving cars (e.g., perception, localization, navigation (route planning) navigation (route planning), maneuver planning, vehicle control, collision avoidance)

Curriculum Revision / Co-Design

1.4 - Robot & Computer Perception (mechanisms of perception - how it makes meaning)

Students should know that

- LO.1.4.6 Edge detection is an early stage of computer perception
 - a. Face Demo - Edge detector

LO.1.4.7 Edge detection involves finding edges in an image as an initial stage to finding more complicated features in images.

LO.1.4.8 Explain how face filters apply graphics to an image

- a. Answer should include facial feature detection and face meshes
- b. Face filters are applying graphics to an image
- c. So they need to know where in the image to apply the graphic and how to distort the graphic to fit the graphic (e.g., size, orientation, skewing) to align with the face
- d. They need to do facial feature detection to identify the places in the image to apply the graphics
- e. They need the mesh to see the 3D structure information to distort the graphic
- f. Limitations of scratch face filters - isn't able to distort the image in 3D. It can only scale and rotate in the plane.

LO.1.4.9 Describe the things self-driving cars and autonomous robots look for

- a. Scene detection, pedestrian intent (body pose)

1.5 - Route Finding